OLS and Linear Algebra: A Gentle Start

A beginner's companion to the *OLS and Linear Algebra* notes

Teaching notes for an introductory machine-learning class

The aim

These notes follow the main *OLS and Linear Algebra* notes chapter for chapter, in the same order, but assume nothing beyond A-level maths. Every symbol is explained the first time it appears, and every chapter has worked examples with small numbers you can check by hand.

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How to use these notes

These notes follow the main *OLS and Linear Algebra* notes chapter for chapter, in the same order, but assume nothing beyond A-level maths. Read a chapter here first; the matching chapter of the main notes will then feel like a summary of something you already know.

- **What you need:** expanding brackets, solving simultaneous equations, and basic differentiation (Chapter 3 recaps it).
- **What you do not need:** any linear algebra. Every symbol is explained the first time it appears.
- **Worked examples** use small numbers you can check by hand. Cover the solution and try each one with a pencil first.
- **Big picture** lines say why a chapter matters for machine learning.

What is OLS, in one paragraph?

Suppose you record how many hours three students studied and the score each got. You want a straight line that predicts score from hours. No line will pass through every point, so each prediction is off by some amount (an *error*). **Ordinary least squares (OLS)** picks the line whose errors, once squared and added up, are as small as possible. Everything in these notes is about doing that neatly for many features at once, using matrices, and understanding when there is exactly one best answer.

Symbols you will meet

Symbol	Read it as	Example
Σ	"add up" (capital sigma)	$\sum_{i=1}^{3} i = 1+2+3 = 6$
$\in$	"is a member of"	$2 \in \{1, 2, 3\}$
$\mathbb{R}$	the real numbers (all numbers on the number line)	$-2,\ 0,\ \pi$ are all in $\mathbb{R}$
$\mathbb{R}^3$	lists of 3 real numbers	$(1, -4, 0.5) \in \mathbb{R}^3$
$\Rightarrow$	"implies"	$x = 2 \Rightarrow x^2 = 4$
$\iff$	"if and only if" (each implies the other)	$x + 1 = 3 \iff x = 2$
$\neq$	"is not equal to"	$3 \neq 4$
$\|v\|$	the length of the vector v	$\|(3, 4)\| = 5$
β	"beta", the usual letter for coefficients	$\beta_0,\ \beta_1$
$\hat{\beta}$	"beta hat": the best (estimated) value of β	$\hat{\beta} = (1, 0.5)$

Note that $\Rightarrow$ only goes one way: $x^2 = 4$ does not imply $x = 2$, because x could be -2.

1 The objects: scalars, vectors, and matrices

Almost everything in OLS is one of three things: a single number, a list of numbers, or a table of numbers.

1.1 Scalars

A **scalar** is one ordinary number, such as 3, -0.5 or π. It has a size but no rows or columns. The name comes from the fact that multiplying by a number *scales* things up or down.

1.2 Vectors

A **vector** is an ordered list of numbers, usually written as a column. For example, a house could be described by (bedrooms, floor area in m^2, age in years):

$$\text{house} = \begin{bmatrix} 3 \\ 85 \\ 20 \end{bmatrix}.$$

Order matters. $(3, 85, 20)$ and $(85, 3, 20)$ are different vectors: the first is a 3-bedroom house, the second an 85-bedroom one. The number of entries is the vector's **dimension**. Writing $\text{house} \in \mathbb{R}^3$ just means "house is a list of 3 real numbers".

In regression, the unknown coefficients are stored in a vector called β:

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} \in \mathbb{R}^3.$$

The small numbers are labels, not powers or multiplication: β_1 is read "beta one" and is simply the second item in the list.

The transpose (written with a small T) turns a column into a row, keeping the entries and their order:

$$a = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \qquad a^{\mathsf{T}} = \begin{bmatrix} 2 & -1 & 5 \end{bmatrix}.$$

Transposing twice gets you back where you started: $(a^{\mathsf{T}})^{\mathsf{T}} = a$.

Two operations you will need constantly.

1. **Adding vectors:** add entry by entry. $(1, 2) + (3, -1) = (4, 1)$. Both vectors must have the same length.
2. **Multiplying by a scalar:** multiply every entry. $3 \times (1, 2) = (3, 6)$.

Combining these gives a **linear combination**: some number of one vector plus some number of another.

Worked example

$2 \times (1, 2) + (-1) \times (3, -1) = (2, 4) + (-3, 1) = (-1, 5)$. Linear combinations are the heart of Chapter 8.

Picture. A vector with 2 entries, such as $(3, 4)$, can be drawn as an arrow from the origin $(0, 0)$ to the point $(3, 4)$. Adding vectors means placing arrows end to end; multiplying by 2 doubles the arrow's length.

1.3 Matrices

A **matrix** is a rectangular table of numbers. Its **size** is written rows × columns, always rows first:

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \end{bmatrix} \quad \text{is } 2 \times 3 \text{ (2 rows, 3 columns).}$$

The entry in row i and column j is written a_{ij}. A matrix with the same number of rows as columns is called **square**.

Worked example

In A above, $a_{12} = 4$ (row 1, column 2) and $a_{23} = 8$ (row 2, column 3).

The transpose of a matrix swaps rows and columns: row 1 becomes column 1, and so on.

$$A^{\mathsf{T}} = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \quad \text{is } 3 \times 2.$$

The design matrix. In regression the data sits in a matrix X, laid out like a spreadsheet: one **row per observation** (a person, a house) and one **column per feature** (hours studied, floor area). With n observations and p features, X is $n \times p$, and x_{ij} is feature j for observation i:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbb{R}^{n \times p}.$$

The domino rule for multiplying. A product AB only makes sense if the number of *columns* of A equals the number of *rows* of B. The two outer numbers give the size of the answer:

$$\underbrace{(m \times n)}_{A} \; \underbrace{(n \times s)}_{B} = \underbrace{(m \times s)}_{AB}.$$

Think of dominoes: the touching ends must match.

Worked examples

- (3×2) times (2×1): inner numbers 2 and 2 match, so the answer is 3×1.
- (1×3) times (3×1): allowed, the answer is 1×1, a single number.
- (2×3) times (2×1): inner numbers 3 and 2 differ, so this product does not exist.

1.4 The regression objects and their dimensions

Here are all the objects used in OLS. Do not worry if the last few rows mean nothing yet; each is built step by step in Chapters 2 and 5.

Object	Shape	Plain-English meaning
X	$n \times p$	The data: one row per observation, one column per feature
β	$p \times 1$	The unknown coefficients we want to find
y	$n \times 1$	The true outcomes we observed
$X\beta$	$n \times 1$	Our predicted outcomes
$e = y - X\beta$	$n \times 1$	The errors (residuals): truth minus prediction
$e^\mathsf{T} e$	1×1	The sum of squared errors: one number
$X^\mathsf{T} X$	$p \times p$	A square table comparing every feature with every other
$X^\mathsf{T} y$	$p \times 1$	A list comparing every feature with the outcomes

Worked example

Three students ($n = 3$). We use $p = 2$ columns: a column of 1s (explained in Chapter 2) and hours studied. Then X is 3×2, β is 2×1 and y is 3×1. By the domino rule, $X\beta$ is $(3 \times 2)(2 \times 1) = 3 \times 1$: one prediction per student, as it should be.

Big picture. keeping track of shapes catches most mistakes before they happen. If the dominoes do not match, the formula is wrong.

2 Matrix multiplication, unpacked

Multiplying a matrix by a vector follows one recipe: for each row, multiply matching entries and add them up.

2.1 Why $X\beta$ gives all predictions

The recipe, with numbers first. Take

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \qquad \beta = \begin{bmatrix} 3 \\ 0.5 \end{bmatrix}.$$

X is 3×2 and β is 2×1, so $X\beta$ will be 3×1. Go row by row:

1. Row 1 is $(1, 2)$. Multiply with $(3, 0.5)$ entry by entry and add: $1 \times 3 + 2 \times 0.5 = 4$.
2. Row 2 is $(1, 4)$: $1 \times 3 + 4 \times 0.5 = 5$.
3. Row 3 is $(1, 5)$: $1 \times 3 + 5 \times 0.5 = 5.5$.

$$X\beta = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 5.5 \end{bmatrix}.$$

What did we just compute? The straight line $y = 3 + 0.5x$, evaluated at $x = 2$, 4 and 5. The second column of X holds the x-values; β holds the intercept (3) and the slope (0.5). One multiplication produced every prediction at once.

Why the column of 1s? It multiplies β_0, so β_0 gets added to every prediction unchanged. That is exactly what an intercept does. Without the 1s the line would be forced through the origin.

The same thing with letters. With $\beta = (\beta_0, \beta_1)$:

$$X\beta = \begin{bmatrix} \beta_0 + 2\beta_1 \\ \beta_0 + 4\beta_1 \\ \beta_0 + 5\beta_1 \end{bmatrix}.$$

In general, entry i of $X\beta$ is row i of X combined with β:

$$(X\beta)_i = \sum_{j=1}^{p} x_{ij}\,\beta_j = x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{ip}\beta_p.$$

The Σ just says "add up this pattern for $j = 1, 2, \ldots, p$". Nothing more.

Matrix times matrix. The same recipe, done for every column of the second matrix. Entry (i, j) of AB is row i of A combined with column j of B.

Worked example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}, \quad AB = \begin{bmatrix} 1\cdot 5 + 2\cdot 7 & 1\cdot 6 + 2\cdot 8 \\ 3\cdot 5 + 4\cdot 7 & 3\cdot 6 + 4\cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Now try BA yourself. You should get $\left[\begin{smallmatrix} 23 & 34 \\ 31 & 46 \end{smallmatrix}\right]$.

So $AB \neq BA$ in general: **order matters**, unlike with ordinary numbers. Keep this in mind whenever we expand brackets.

2.2 Dot products and squared length

The **dot product** of two vectors of the same length multiplies matching entries and adds them up. Written with the transpose, it is a row times a column, $(1 \times n)(n \times 1) = 1 \times 1$, a single number:

$$a^{\mathsf{T}} b = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

Worked example

$a = (1, 2, 3)$, $b = (4, -1, 2)$. Then $a^{\mathsf{T}} b = 4 - 2 + 6 = 8$.

A vector dotted with itself gives its squared length. For $a = (3, 4)$: $a^{\mathsf{T}} a = 3^2 + 4^2 = 25$, so its length is $\sqrt{25} = 5$. This is just Pythagoras: the arrow to $(3, 4)$ is the hypotenuse of a 3-4-5 triangle.

$$a^{\mathsf{T}} a = a_1^2 + a_2^2 + a_3^2 = \|a\|^2 \geq 0.$$

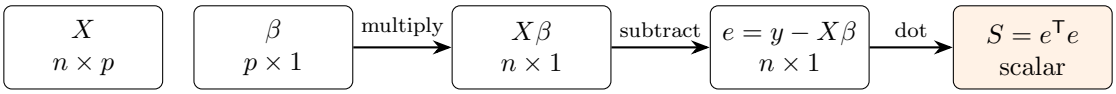
Two facts follow from "it is a sum of squares", and both are used again in Chapter 7:

- It can **never be negative**, because squares are never negative.
- It equals **zero only when every entry is zero**. If even one entry is non-zero, its square is positive.

Perpendicular vectors. If $a^{\mathsf{T}} b = 0$ the vectors are at right angles. For example, $(1, 2)$ and $(2, -1)$ give $2 - 2 = 0$. Sketch them: they meet at $90°$.

2.3 A visual map of the OLS dimensions

The whole OLS objective is built in three steps. Each step uses only the recipes above.



1. **Predict.** X $(n \times p)$ times β $(p \times 1)$ gives $X\beta$ $(n \times 1)$: one prediction per observation.
2. **Compare.** Subtract from the truth: $e = y - X\beta$ $(n \times 1)$, one error per observation.
3. **Score.** Dot e with itself: $S = e^\mathsf{T} e$, a single number, the sum of squared errors.

Worked example

Use X and $\beta = (3, 0.5)$ from Section 2.1, and suppose the true scores were $y = (4, 6, 5)$.

1. Predictions: $X\beta = (4, 5, 5.5)$.
2. Errors: $e = (4 - 4,\ 6 - 5,\ 5 - 5.5) = (0, 1, -0.5)$.
3. Score: $S = 0^2 + 1^2 + (-0.5)^2 = 1.25$.

A different β would give a different S. **OLS is the search for the β that makes S as small as possible.**

3 Derivatives: the scalar rules first

A derivative measures how fast something changes; to find the bottom of a valley, we look for where the slope is zero.

3.1 Derivative of a scalar function

For a function $f(\beta)$ of one variable, the derivative $f'(\beta)$ is the gradient of its graph: how much f changes per small nudge in β. At the lowest point of a smooth U-shaped curve the graph is flat, so $f'(\beta) = 0$ **there**. That single idea drives all of OLS.

The rules you need, all from A-level:

Rule	In words	Example
$\frac{d}{d\beta} c = 0$	constants do not change	$\frac{d}{d\beta} 7 = 0$
$\frac{d}{d\beta}(a\beta) = a$	a straight line has constant slope	$\frac{d}{d\beta}(5\beta) = 5$
$\frac{d}{d\beta}\beta^n = n\beta^{n-1}$	power rule	$\frac{d}{d\beta}\beta^2 = 2\beta$
$\frac{d}{d\beta} f(g(\beta)) = f'(g(\beta))\, g'(\beta)$	chain rule: outside times inside	see below

Worked example 1

$f(\beta) = 3\beta^2 + 5\beta + 7$. Differentiate term by term: $f'(\beta) = 6\beta + 5 + 0 = 6\beta + 5$.

Worked example 2: the chain rule, and our first tiny OLS

One data point: $x = 2$, $y = 6$. We predict y with the line $x\beta$ through the origin and want the β that makes the squared error smallest:

$$f(\beta) = (y - x\beta)^2 = (6 - 2\beta)^2.$$

1. **Outside:** the function is "something squared", u^2, whose derivative is $2u$.
2. **Inside:** $u = 6 - 2\beta$, whose derivative is -2.
3. **Multiply:** $f'(\beta) = 2(6 - 2\beta) \times (-2) = -4(6 - 2\beta)$.

Check by expanding instead: $(6 - 2\beta)^2 = 36 - 24\beta + 4\beta^2$, whose derivative is $-24 + 8\beta = -4(6 - 2\beta)$. Same answer.
Set it to zero: $-4(6 - 2\beta) = 0$ gives $\beta = 3$. The line $y = 3x$ passes exactly through $(2, 6)$, so the error is zero: clearly the best possible. It is a minimum, not a maximum, because f is a U-shaped parabola (the β^2 term, $4\beta^2$, has a positive coefficient).

The general version is exactly the same calculation with letters: $f(\beta) = (y - x\beta)^2$ gives $f'(\beta) = 2(y - x\beta)(-x)$. The $-x$ is the derivative of the inside.

3.2 Several parameters: partial derivatives

When a function has two unknowns, β_1 and β_2, we ask two separate questions: how does f change if I nudge only β_1? And if I nudge only β_2? Each answer is a **partial derivative**, written with a curly ∂ ("partial"). The rule: **treat every other letter as a constant**, exactly as you treated x and y as constants in Section 3.1.

Worked example 1

$f(\beta_1, \beta_2) = \beta_1^2 + 3\beta_1\beta_2 + \beta_2^2$.

- $\partial f/\partial\beta_1$: treat β_2 as a number. β_1^2 gives $2\beta_1$; $3\beta_2 \times \beta_1$ gives $3\beta_2$; β_2^2 is a constant, giving 0. So $\partial f/\partial\beta_1 = 2\beta_1 + 3\beta_2$.
- $\partial f/\partial\beta_2$: treat β_1 as a number. Similarly $\partial f/\partial\beta_2 = 3\beta_1 + 2\beta_2$.

Worked example 2

$f(\beta_1, \beta_2) = (y - \beta_1 - 2\beta_2)^2$. Chain rule each time; only the inside derivative changes:

$$\frac{\partial f}{\partial \beta_1} = 2(y - \beta_1 - 2\beta_2)(-1), \qquad \frac{\partial f}{\partial \beta_2} = 2(y - \beta_1 - 2\beta_2)(-2) = -4(y - \beta_1 - 2\beta_2).$$

The gradient is all the partial derivatives stacked into one column vector, written ∇f ("grad f"):

$$\nabla_\beta f = \begin{bmatrix} \partial f/\partial\beta_1 \\ \vdots \\ \partial f/\partial\beta_p \end{bmatrix} \in \mathbb{R}^p.$$

For worked example 1, $\nabla f = (2\beta_1 + 3\beta_2,\ 3\beta_1 + 2\beta_2)$. At the point $\beta = (1, 1)$ this is $(5, 5)$. There is nothing new here: a gradient is just a list of ordinary derivatives, one per parameter.

3.3 What is a derivative with respect to a vector?

"Differentiate with respect to the vector β" just means "differentiate with respect to each of its entries". What you get back depends on what the function **outputs**:

Function outputs...	You get...	Shape
one number, $f(\beta)$	the **gradient**: one partial derivative per parameter	$p \times 1$
a list of m numbers, $g(\beta)$	the **Jacobian**: a table, row i = the gradient of output i	$m \times p$
a whole matrix	a 3-way stack of derivatives (a tensor)	not needed here

Worked example: a Jacobian

Take $g(\beta) = X\beta$ with the X from Chapter 2. Its three outputs are $g_1 = \beta_0 + 2\beta_1$, $g_2 = \beta_0 + 4\beta_1$, $g_3 = \beta_0 + 5\beta_1$. Differentiating each:

$$J = \begin{bmatrix} \partial g_1/\partial\beta_0 & \partial g_1/\partial\beta_1 \\ \partial g_2/\partial\beta_0 & \partial g_2/\partial\beta_1 \\ \partial g_3/\partial\beta_0 & \partial g_3/\partial\beta_1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} = X.$$

The Jacobian of $X\beta$ is X itself, just as the derivative of 5β is 5.

Big picture. the OLS score $S(\beta)$ outputs one number, so its derivative is a gradient: a list of p ordinary derivatives. X and y are fixed data, treated like the constants 2 and 6 in Section 3.1. We never differentiate the data.

4 OLS first in scalar form

Before any matrices, we fit a line by hand: write the squared-error total, differentiate, set to zero, solve. The answer for our running example is the line $\hat{y} = 1 + 0.5x$.

The running example (used again in Chapters 5, 7 and 10):

Student	Hours studied, x	Score, y
1	1	1
2	2	3
3	3	2

Step 1: predictions and errors. For a line with intercept β_0 and slope β_1, student i gets prediction $\hat{y}_i = \beta_0 + \beta_1 x_i$ ($\hat{y}$, "y hat", means *predicted* y). The **residual** (error) is truth minus prediction: $e_i = y_i - \hat{y}_i$.

Step 2: why square the errors? Adding raw errors fails: errors of $+2$ and -2 add to 0, as if the fit were perfect. Squaring makes every error count positively ($4 + 4 = 8$). It also punishes big misses more: an error of 3 costs 9, three times as much as three errors of 1.

Step 3: the total squared error.

$$S(\beta_0, \beta_1) = \sum_{i=1}^{3} (y_i - \beta_0 - \beta_1 x_i)^2 = (1 - \beta_0 - \beta_1)^2 + (3 - \beta_0 - 2\beta_1)^2 + (2 - \beta_0 - 3\beta_1)^2.$$

Step 4: differentiate. Use the chain rule on each bracket. With respect to β_0, each inside derivative is -1; with respect to β_1, it is $-x_i$ (that is -1, -2, -3):

$$\frac{\partial S}{\partial \beta_0} = -2\big[(1 - \beta_0 - \beta_1) + (3 - \beta_0 - 2\beta_1) + (2 - \beta_0 - 3\beta_1)\big] = -2\,[\,6 - 3\beta_0 - 6\beta_1\,],$$
$$\frac{\partial S}{\partial \beta_1} = -2\big[1(1 - \beta_0 - \beta_1) + 2(3 - \beta_0 - 2\beta_1) + 3(2 - \beta_0 - 3\beta_1)\big] = -2\,[\,13 - 6\beta_0 - 14\beta_1\,].$$

Step 5: set both to zero (the bottom of the valley is flat in every direction). Dividing by -2 gives two simultaneous equations:

$$3\beta_0 + 6\beta_1 = 6, \qquad 6\beta_0 + 14\beta_1 = 13.$$

Step 6: solve. From the first, $\beta_0 = 2 - 2\beta_1$. Substitute: $6(2-2\beta_1)+14\beta_1 = 13$, so $12+2\beta_1 = 13$, giving $\boldsymbol{\beta_1 = 0.5}$ and $\boldsymbol{\beta_0 = 1}$. The best line is $\hat{y} = 1 + 0.5x$.

Step 7: check. Predictions are 1.5, 2, 2.5, so the residuals are $e = (-0.5, 1, -0.5)$ and $S = 0.25 + 1 + 0.25 = 1.5$. Try any other line (for example $\hat{y} = 2$, a flat line) and you will get a bigger S (here 2).

The general pattern

With p features and n observations, the prediction for observation i is $\hat{y}_i = x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{ip}\beta_p$, and

$$S(\beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \Big(y_i - \sum_{j=1}^{p} x_{ij}\beta_j\Big)^2.$$

Differentiating with respect to one coefficient β_k, the inside derivative is $-x_{ik}$ (only the β_k term moves), so

$$\frac{\partial S}{\partial \beta_k} = -2\sum_{i=1}^{n} x_{ik}\Big(y_i - \sum_{j=1}^{p} x_{ij}\beta_j\Big) = 0, \qquad k = 1, \ldots, p.$$

That is p equations for p unknowns. (Labels only: in the example we called the coefficients β_0 and β_1; in general formulas they are β_1 to β_p.)

What the equations are really saying

Each equation says: **feature column k, dotted with the residuals, equals zero.** Check it on the example, where $e = (-0.5, 1, -0.5)$:

- The column of 1s: $-0.5 + 1 - 0.5 = 0$. (The errors balance out.)
- The hours column $(1, 2, 3)$: $-0.5 + 2 - 1.5 = 0$.

In Chapter 2 we saw a zero dot product means "perpendicular". So at the best fit, the residuals are perpendicular to every feature column. In matrix notation, all p equations become one line: $X^{\mathsf{T}}(y - X\hat{\beta}) = 0$. Chapter 5 derives that line directly.

5 OLS in matrix form, with every step explained

This chapter redoes Chapter 4 in matrix language and lands on the same answer, $\hat{\beta} = (1, 0.5)$. At every step we check the matrix version against the numbers we already have.

The running example in matrix form:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \qquad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

5.1 Step 1: stack the predictions and residuals

$$\hat{y} = X\beta = \begin{bmatrix} \beta_0 + \beta_1 \\ \beta_0 + 2\beta_1 \\ \beta_0 + 3\beta_1 \end{bmatrix}, \qquad e = y - X\beta = \begin{bmatrix} 1 - \beta_0 - \beta_1 \\ 3 - \beta_0 - 2\beta_1 \\ 2 - \beta_0 - 3\beta_1 \end{bmatrix}.$$

These are exactly the three brackets from Chapter 4, stacked in a column.

5.2 Step 2: write the sum of squares as a product

A row times a column is a dot product (Chapter 2), and a vector dotted with itself is the sum of its squared entries:

$$S(\beta) = e^{\mathsf{T}} e = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \sum_{i=1}^{n} e_i^2 = (y - X\beta)^{\mathsf{T}}(y - X\beta).$$

Shapes: $(1 \times n)(n \times 1) = 1 \times 1$. The score is one number, as it should be.

5.3 Step 3: expand using transpose rules

Two rules are needed:

$$(A - B)^{\mathsf{T}} = A^{\mathsf{T}} - B^{\mathsf{T}}, \qquad (AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}.$$

The first is obvious: flipping a difference flips each part. The second **reverses the order**. Think of socks and shoes: you put socks on then shoes, but take shoes off then socks.

Worked example: checking $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$

$A = \begin{bmatrix} 1 & 2 \end{bmatrix}$ (1×2) and $B = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then $AB = 3 + 8 = 11$, and $B^{\mathsf{T}} A^{\mathsf{T}} = \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3 + 8 = 11$. Same. (In the original order, A^{T} is 2×1 and B^{T} is 1×2, so $A^{\mathsf{T}} B^{\mathsf{T}}$ would be a 2×2 table, not even the right shape.)

Now expand like $(a - b)(c - d) = ac - ad - bc + bd$, but **keep the order of every product**, since $AB \neq BA$:

$$\begin{aligned} S(\beta) &= (y - X\beta)^{\mathsf{T}}(y - X\beta) \\ &= (y^{\mathsf{T}} - \beta^{\mathsf{T}} X^{\mathsf{T}})(y - X\beta) \\ &= y^{\mathsf{T}} y - y^{\mathsf{T}} X\beta - \beta^{\mathsf{T}} X^{\mathsf{T}} y + \beta^{\mathsf{T}} X^{\mathsf{T}} X\beta. \end{aligned}$$

The two middle terms are equal. Each is a single number, and a single number transposed is itself. So $y^\mathsf{T} X\beta = (y^\mathsf{T} X\beta)^\mathsf{T} = \beta^\mathsf{T} X^\mathsf{T} y$.

Worked example: the middle terms with numbers

$X^\mathsf{T} y = (1+3+2,\ 1+6+6) = (6, 13)$, so $\beta^\mathsf{T} X^\mathsf{T} y = 6\beta_0 + 13\beta_1$. And $y^\mathsf{T} X\beta = 1(\beta_0+\beta_1) + 3(\beta_0+2\beta_1) + 2(\beta_0+3\beta_1) = 6\beta_0 + 13\beta_1$. Equal.

Therefore:

$$S(\beta) = y^\mathsf{T} y - 2\beta^\mathsf{T} X^\mathsf{T} y + \beta^\mathsf{T} X^\mathsf{T} X\beta.$$

Meet $X^\mathsf{T} X$. Entry (j, k) of $X^\mathsf{T} X$ is column j of X dotted with column k. For our X, the columns are $c_1 = (1, 1, 1)$ and $c_2 = (1, 2, 3)$:

$$X^\mathsf{T} X = \begin{bmatrix} c_1 \cdot c_1 & c_1 \cdot c_2 \\ c_2 \cdot c_1 & c_2 \cdot c_2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}.$$

It is **symmetric** (equal to its own transpose) because $c_1 \cdot c_2 = c_2 \cdot c_1$. In symbols: $(X^\mathsf{T} X)^\mathsf{T} = X^\mathsf{T}(X^\mathsf{T})^\mathsf{T} = X^\mathsf{T} X$.

Worked example: the full objective

With $y^\mathsf{T} y = 1 + 9 + 4 = 14$:

$$S(\beta) = 14 - 2(6\beta_0 + 13\beta_1) + (3\beta_0^2 + 12\beta_0\beta_1 + 14\beta_1^2).$$

If you expand the three brackets from Chapter 4 by hand, you get exactly this.

5.4 Step 4: derivative rules for vectors and matrices

We need the gradient of each piece. Each rule below is checked with $p = 2$ by writing everything out.

Rule 1: constants vanish. $y^\mathsf{T} y$ contains no β, so its gradient is 0.

Rule 2: $\nabla(a^\mathsf{T}\beta) = a$. Write $a^\mathsf{T}\beta = a_1\beta_1 + a_2\beta_2$. Its partial derivatives are a_1 and a_2, so the gradient is a. Since $\beta^\mathsf{T} a$ is the same number, $\nabla(\beta^\mathsf{T} a) = a$ too. *Example:* $\beta^\mathsf{T} X^\mathsf{T} y = 6\beta_0 + 13\beta_1$ has gradient $(6, 13) = X^\mathsf{T} y$.

Rule 3: $\nabla(\beta^\mathsf{T} A\beta) = (A + A^\mathsf{T})\beta$, which is $2A\beta$ when A is symmetric. Write it out for a 2×2 matrix:

$$\beta^\mathsf{T} A\beta = a_{11}\beta_1^2 + (a_{12} + a_{21})\beta_1\beta_2 + a_{22}\beta_2^2.$$

Differentiating with respect to β_1 gives $2a_{11}\beta_1 + (a_{12}+a_{21})\beta_2$, which is the first entry of $(A+A^\mathsf{T})\beta$. Why both a_{12} and a_{21}? Because β_1 appears on the left of some products and on the right of others, and both count.

Worked example: Rule 3, not symmetric

$A = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right]$. Then $\beta^\mathsf{T} A\beta = \beta_1^2 + 2\beta_1\beta_2 + 3\beta_2^2$, with gradient $(2\beta_1 + 2\beta_2,\ 2\beta_1 + 6\beta_2)$. And $A + A^\mathsf{T} = \left[\begin{smallmatrix} 2 & 2 \\ 2 & 6 \end{smallmatrix}\right]$, so $(A + A^\mathsf{T})\beta$ gives the same. Using $2A\beta$ here would be wrong.

Worked example: Rule 3, symmetric (our $X^\mathsf{T} X$)

$\beta^\mathsf{T} X^\mathsf{T} X\beta = 3\beta_0^2 + 12\beta_0\beta_1 + 14\beta_1^2$ has gradient $(6\beta_0 + 12\beta_1,\ 12\beta_0 + 28\beta_1) = 2X^\mathsf{T} X\beta$.

Put the pieces together:

$$\nabla_\beta S(\beta) = 0 - 2X^\mathsf{T} y + 2X^\mathsf{T} X\beta.$$

Check: $-2(6, 13) + 2(3\beta_0 + 6\beta_1,\ 6\beta_0 + 14\beta_1) = -2(6 - 3\beta_0 - 6\beta_1,\ 13 - 6\beta_0 - 14\beta_1)$. These are exactly the two partial derivatives from Chapter 4.

5.5 Step 5: set the gradient equal to zero

At the minimum every partial derivative is zero. Divide by 2 and rearrange to get the **normal equations**:

$$\boxed{X^\mathsf{T} X\hat{\beta} = X^\mathsf{T} y}$$

For our example this reads $\left[\begin{smallmatrix} 3 & 6 \\ 6 & 14 \end{smallmatrix}\right]\hat{\beta} = \left[\begin{smallmatrix} 6 \\ 13 \end{smallmatrix}\right]$, which is the pair $3\beta_0 + 6\beta_1 = 6$ and $6\beta_0 + 14\beta_1 = 13$ from Chapter 4. Row k of the normal equations is "$\partial S/\partial\beta_k = 0$".

Why "normal"? In geometry, *normal* means perpendicular. Rearranged, the equations say $X^\mathsf{T}(y - X\hat{\beta}) = 0$: the residuals are perpendicular to every column of X, just as we checked at the end of Chapter 4.

5.6 Step 6: solve when the inverse exists

You cannot divide by a matrix. Instead, if $X^\mathsf{T} X$ has an **inverse** $(X^\mathsf{T} X)^{-1}$, the matrix version of $1/a$, multiply both sides on the left by it:

$$\boxed{\hat{\beta} = (X^\mathsf{T} X)^{-1} X^\mathsf{T} y}$$

For a 2×2 matrix there is a formula you can memorise. The number $ad - bc$ is called the **determinant**:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Worked example

For $X^\mathsf{T} X = \left[\begin{smallmatrix} 3 & 6 \\ 6 & 14 \end{smallmatrix}\right]$, the determinant is $3 \times 14 - 6 \times 6 = 6$. So

$$\hat{\beta} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 13 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 84 - 78 \\ -36 + 39 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}.$$

The same line as Chapter 4: $\hat{y} = 1 + 0.5x$.

The catch: the formula divides by the determinant. If $ad - bc = 0$ there is no inverse, just as $1/0$ does not exist. Chapters 6 to 9 explain exactly when that happens and what it means.

6 Inequalities and definiteness: the vocabulary

"Positive definite" and "positive semidefinite" are the matrix versions of "positive" and "non-negative", and they tell us whether the OLS valley has one lowest point.

Symbol	Meaning	Example
$a > b$	a is strictly bigger than b	$5 > 3$ true; $3 > 3$ false
$a \geq b$	a is bigger than or equal to b	$3 \geq 3$ true
$a = b$ (vectors)	every matching entry is equal	$(1,2) = (1,2)$; $(1,2) \neq (2,1)$
$a^\mathsf{T} b = 0$	a and b are perpendicular (orthogonal)	$(1,2)$ and $(2,-1)$
$A \succeq 0$	positive semidefinite: $v^\mathsf{T} A v \geq 0$ for every v	$\left[\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right]$ (below)
$A \succ 0$	positive definite: $v^\mathsf{T} A v > 0$ for every $v \neq 0$	$\left[\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right]$ (below)

6.1 Positive semidefinite versus positive definite

The quadratic form. Given a symmetric matrix A and any vector v, the expression $v^\mathsf{T} A v$ is one number (shapes: $(1 \times p)(p \times p)(p \times 1) = 1 \times 1$). For a 2×2 symmetric matrix $A = \left[\begin{smallmatrix}a&b\\b&d\end{smallmatrix}\right]$ and $v = (v_1, v_2)$, it works out as

$$v^\mathsf{T} A v = a\, v_1^2 + 2b\, v_1 v_2 + d\, v_2^2.$$

Start with 1×1 matrices. If A is just the number $[a]$, then $v^\mathsf{T} A v = a v^2$. This is positive for every $v \neq 0$ exactly when $a > 0$, and never negative exactly when $a \geq 0$. So:

- **Positive definite** ($A \succ 0$) is the matrix version of "$a > 0$": $v^\mathsf{T} A v > 0$ for every non-zero v.
- **Positive semidefinite** ($A \succeq 0$) is the matrix version of "$a \geq 0$": $v^\mathsf{T} A v \geq 0$ for every v, and zero is allowed even when $v \neq 0$.

The only difference is whether a non-zero v is allowed to give exactly 0.

Worked example 1: positive definite

$A = \left[\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right]$. Then $v^\mathsf{T} A v = v_1^2 + v_2^2$. This is zero only when $v_1 = v_2 = 0$, so it is positive for every non-zero v.

Worked example 2: semidefinite but not definite

$A = \left[\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right]$. Then $v^\mathsf{T} A v = v_1^2 + 2v_1 v_2 + v_2^2 = (v_1 + v_2)^2$. A square is never negative, so $A \succeq 0$. But $v = (1, -1)$ is not zero and gives $(1 - 1)^2 = 0$, so A is **not** positive definite.

Worked example 3: neither

$A = \left[\begin{smallmatrix}1&0\\0&-1\end{smallmatrix}\right]$. Then $v^\mathsf{T} A v = v_1^2 - v_2^2$. For $v = (0, 1)$ this is -1, which is negative.

Picture it as a surface. Plot $z = v^\mathsf{T} A v$ over the (v_1, v_2) plane:

- Positive definite: a **bowl**. It curves up in every direction, with one lowest point.
- Positive semidefinite (not definite): a **trough** or half-pipe. It curves up in some directions but has a flat floor along one line, so there is a whole line of lowest points.
- Neither: a **saddle** (a crisp shape), up in one direction and down in another.

Keep the bowl and the trough in mind: in Chapter 10 the OLS score $S(\beta)$ turns out to be one of these two shapes.

7 Why $X^\mathsf{T}X$ is positive semidefinite

For any data matrix X, $v^\mathsf{T}X^\mathsf{T}Xv$ is a squared length, so it can never be negative. That one line is the whole proof.

First, $X^\mathsf{T}X$ is symmetric (Chapter 5): entry (j,k) is column j dotted with column k, which equals column k dotted with column j.

The key trick. Using $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$, we have $(Xv)^\mathsf{T} = v^\mathsf{T}X^\mathsf{T}$. So we can regroup:

$$v^\mathsf{T}X^\mathsf{T}Xv = (Xv)^\mathsf{T}(Xv) = \|Xv\|^2 \geq 0.$$

In words: multiply v by X to get a new vector Xv, then take its squared length. Squared lengths are sums of squares (Chapter 2), so they are never negative. Hence $\boldsymbol{X^\mathsf{T}X \succeq 0}$ **for every** X **whatsoever**, including badly behaved ones.

Worked example

Use the running example $X = \left[\begin{smallmatrix}1&1\\1&2\\1&3\end{smallmatrix}\right]$, with $X^\mathsf{T}X = \left[\begin{smallmatrix}3&6\\6&14\end{smallmatrix}\right]$, and pick $v = (2,-1)$.

- **Route 1 (via Xv):** $Xv = (2-1,\ 2-2,\ 2-3) = (1,0,-1)$. Squared length: $1+0+1 = 2$.
- **Route 2 (via $X^\mathsf{T}X$):** $v^\mathsf{T}X^\mathsf{T}Xv = 3(2)^2 + 2\times 6\times(2)(-1) + 14(-1)^2 = 12-24+14 = 2$.

Both routes give 2, as the trick promises. Try other v's: you will never get a negative answer.

7.1 When is it positive definite?

Positive definite needs $v^\mathsf{T}X^\mathsf{T}Xv > 0$ for every non-zero v. Since this number is $\|Xv\|^2$, and a squared length is zero only for the zero vector, we need:

$$Xv \neq 0 \quad \text{for every } v \neq 0.$$

What does $Xv = 0$ mean? Xv is a linear combination of the columns of X, with the entries of v as the weights (Chapter 2's recipe, read column by column):

$$Xv = v_1\begin{bmatrix}x_{11}\\ \vdots \\ x_{n1}\end{bmatrix} + v_2\begin{bmatrix}x_{12}\\ \vdots \\ x_{n2}\end{bmatrix} + \cdots + v_p\begin{bmatrix}x_{1p}\\ \vdots \\ x_{np}\end{bmatrix}.$$

So $Xv = 0$ with $v \neq 0$ means **some mixture of the columns cancels out completely**. Columns that can do this are called linearly dependent (Chapter 8).

Worked example 1: nothing cancels

For the running example, $Xv = v_1(1,1,1) + v_2(1,2,3) = (v_1+v_2,\ v_1+2v_2,\ v_1+3v_2)$. Setting this to zero: the first two entries give $v_1+v_2 = 0$ and $v_1+2v_2 = 0$. Subtracting, $v_2 = 0$, and then $v_1 = 0$. Only $v = 0$ works, so $X^\mathsf{T}X$ is positive definite.

Worked example 2: something cancels

$X = \left[\begin{smallmatrix}1&2\\2&4\\3&6\end{smallmatrix}\right]$. The second column is twice the first, so $2\times$ (column 1) $-\ 1\times$ (column 2) $= 0$. In other words $v = (2,-1)$, or equally $(-2,1)$, gives $Xv = 0$. So $X^\mathsf{T}X$ is only semidefinite.

The chain of equivalent statements ($\iff$ means "if and only if": each statement guarantees the next and vice versa):

$$\begin{aligned} X \text{ has full column rank} &\iff \text{Null}(X) = \{0\} \\ &\iff X^\mathsf{T} X \succ 0 \\ &\iff X^\mathsf{T} X \text{ is invertible.} \end{aligned}$$

The terms "rank" and "Null" are defined in Chapters 8 and 9; for now read the first two lines as "no mixture of the columns cancels out".

8 Rank and linear independence

The rank of a matrix counts how many of its columns carry genuinely new information; OLS needs every column to count.

8.1 What does rank mean?

Linear independence. A set of vectors is **linearly independent** if none of them can be built as a linear combination of the others. An equivalent test, often easier to use: the only way to mix them into the zero vector is to use zero of each.

$$c_1 a_1 + c_2 a_2 + \cdots + c_p a_p = 0 \;\Rightarrow\; c_1 = c_2 = \cdots = c_p = 0.$$

If some non-zero mixture gives zero, the vectors are **linearly dependent**: at least one of them is redundant.

Worked examples

Vectors	Independent?	Why
$(1,0)$ and $(0,1)$	Yes	$c_1(1,0)+c_2(0,1) = (c_1,c_2)$, zero only if $c_1 = c_2 = 0$
$(1,2)$ and $(2,4)$	No	$(2,4) = 2 \times (1,2)$, so $2(1,2) - (2,4) = 0$
$(1,0)$, $(0,1)$, $(1,1)$	No	$(1,1) = (1,0) + (0,1)$
$(1,0,1)$, $(0,1,1)$, $(1,1,2)$	No	the third is the sum of the first two
$(1,1,1)$ and $(1,2,3)$	Yes	shown in Chapter 7, worked example 1

Notice the third row: in 2-dimensional space you can never have more than 2 independent vectors. There are only two directions to go in.

Rank. The **rank** of a matrix is the largest number of linearly independent columns it has: the number of genuinely different directions among its columns. (A theorem says counting rows instead gives the same number.)

Worked example

The matrix $\left[\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right]$ has columns $(1,0)$, $(0,1)$, $(1,1)$. The first two are independent and the third is their sum, so the rank is 2.

Full column rank. An $n \times p$ matrix X has **full column rank** when $\text{rank}(X) = p$, meaning every column brings something new and none is redundant.

Machine-learning examples of dependent columns:

- Height in centimetres and height in metres: one column is 100 times the other.
- The **dummy variable trap**: an intercept column of 1s, a column "is_male" (1 or 0) and a column "is_female" (1 or 0). For every person is_male + is_female = 1, so is_male + is_female − intercept = 0.
- Total spend, plus spend on food, plus spend on everything else: the last two add up to the first.

8.2 A visual example of dependent columns

Take

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

The second column is twice the first, so there is only one genuine direction: $\text{rank}(X) = 1$, less than $p = 2$. Picture each column as an arrow in 3D: both arrows point the same way along one line.

The mixture $v = (-2, 1)$ cancels out:

$$Xv = \begin{bmatrix} -2+2 \\ -4+4 \\ -6+6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now compute $X^\mathsf{T}X$ (column dot products: $1+4+9 = 14$, $2+8+18 = 28$, $4+16+36 = 56$):

$$X^\mathsf{T}X = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}, \qquad \det = 14 \times 56 - 28 \times 28 = 784 - 784 = 0.$$

The determinant is zero, so the 2×2 inverse formula from Chapter 5 would divide by zero. We can also see directly that $X^\mathsf{T}X$ sends v to zero: $X^\mathsf{T}Xv = (14\times(-2)+28,\ 28\times(-2)+56) = (0, 0)$. In general, $Xv = 0$ implies $X^\mathsf{T}Xv = X^\mathsf{T}0 = 0$.

What it means for a model: if column 2 is just column 1 doubled (say, price in pounds and price in half-pounds), the data cannot tell how to split the credit between them. That is the whole problem.

8.3 Why square matrices and inverses matter

Only **square** matrices can have an ordinary inverse (Chapter 9 explains why). A design matrix is usually tall: many more observations than features, say $1{,}000 \times 5$. So X itself has no inverse. But $X^\mathsf{T}X$ is always $p \times p$, which is square, so it is the natural candidate to invert.

A square matrix is invertible exactly when its columns are linearly independent. And a useful fact, proved in Section 9.7, links $X^\mathsf{T}X$ back to X:

$$\text{rank}(X^\mathsf{T}X) = \text{rank}(X).$$

Worked example: a comparison

X	rank(X)	$X^\top X$	$\det(X^\top X)$	Invertible?
$\left[\begin{smallmatrix}1&1\\1&2\\1&3\end{smallmatrix}\right]$ (running example)	2	$\left[\begin{smallmatrix}3&6\\6&14\end{smallmatrix}\right]$	6	Yes
$\left[\begin{smallmatrix}1&2\\2&4\\3&6\end{smallmatrix}\right]$ (dependent)	1	$\left[\begin{smallmatrix}14&28\\28&56\end{smallmatrix}\right]$	0	No

So $X^\top X$ is invertible exactly when X has full column rank.

9 Inverses, bijections, and rank: how it all connects

A square matrix has an inverse exactly when every output comes from exactly one input; the only way a matrix can fail this is by squashing some non-zero direction down to zero.

This chapter explains why words like invertible, singular, null space, rank and positive definite keep appearing together. We start from the very idea of a function.

The big idea

A square matrix has an inverse exactly when the map $v \mapsto Av$ is a *bijection*: every output comes from exactly one input. For a matrix, the only way this can fail is that some non-zero direction is collapsed to 0. A collapsed direction is also exactly what separates positive semidefinite from positive definite.

9.1 The definition of a matrix inverse

Start with numbers. The inverse of 5 is 1/5, because $5 \times 1/5 = 1$. The number 1 is special: multiplying by it changes nothing. The number 0 has no inverse, since $0 \times$ anything $= 0$, never 1.

The matrix version of 1 is the **identity matrix** I: ones down the diagonal, zeros everywhere else. Multiplying by it changes nothing:

$$I_2 = \begin{bmatrix}1&0\\0&1\end{bmatrix}, \qquad I_2 \begin{bmatrix}3\\7\end{bmatrix} = \begin{bmatrix}1\cdot 3 + 0\cdot 7\\0\cdot 3 + 1\cdot 7\end{bmatrix} = \begin{bmatrix}3\\7\end{bmatrix}.$$

Definition. A square $p \times p$ matrix A is **invertible** if there is a matrix A^{-1} with

$$AA^{-1} = I_p \quad \text{and} \quad A^{-1}A = I_p.$$

A square matrix with no inverse is called **singular** (the matrix version of 0).

Worked example

Is $\left[\begin{smallmatrix}5&-3\\-3&2\end{smallmatrix}\right]$ the inverse of $A = \left[\begin{smallmatrix}2&3\\3&5\end{smallmatrix}\right]$? Multiply, row by column:

$$\begin{bmatrix}2&3\\3&5\end{bmatrix}\begin{bmatrix}5&-3\\-3&2\end{bmatrix} = \begin{bmatrix}10-9&-6+6\\15-15&-9+10\end{bmatrix} = \begin{bmatrix}1&0\\0&1\end{bmatrix}.$$

Yes. (Multiplying in the other order also gives I; try it.)

An inverse, if it exists, is unique. Suppose both B and C undo A: $AB = I$ and $CA = I$. Then, using only "multiplying by I changes nothing" and the fact that brackets can be regrouped:

$$C = CI = C(AB) = (CA)B = IB = B.$$

So B and C were the same matrix all along. That is why we can say *the* inverse.

9.2 Matrices as functions

A function is a machine. You feed in an input, it gives back exactly one output. For example $f(x) = 2x + 1$ turns 3 into 7. Two words describe where things live:

- The **domain** is the set of allowed inputs.
- The **codomain** is the set where outputs are allowed to land. Not every point of the codomain has to actually be produced.

A matrix is a machine too. An $m \times p$ matrix A takes a vector v with p entries and returns Av with m entries:

$$v \longmapsto Av, \qquad \underbrace{\mathbb{R}^p}_{\text{inputs}} \to \underbrace{\mathbb{R}^m}_{\text{outputs}}.$$

The number of **columns** fixes the inputs (by the domino rule, v must have p entries); the number of **rows** fixes the outputs.

Worked example

$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ is 3×2, so it turns 2-vectors into 3-vectors. With $v = (1, 1)$: $Av = (1 + 2,\ 0 + 1,\ 1 + 0) = (3, 1, 1)$. For a design matrix, X turns a coefficient vector β (p entries) into a prediction vector $X\beta$ (n entries).

Multiplying matrices = running machines one after another. $A(Bv) = (AB)v$: "do B, then do A" is the same as the single machine AB. The identity I is the "do nothing" machine. So the definition of the inverse says:

- $A^{-1}A = I$: do A, then A^{-1}, and you get back what you started with. A^{-1} is the **undo** button for A.
- $AA^{-1} = I$: do A^{-1}, then A, and you also get back what you started with.

9.3 Injective, surjective, and bijective functions

When can a machine be undone? It depends on two properties. Here is a story to remember them: inputs are **people**, outputs are **seats** in a theatre, and the function tells each person which seat to sit in.

Property	Definition	Theatre version	Test
Injective (one-to-one)	different inputs always give different outputs	nobody shares a seat (some seats may stay empty)	if $f(a) = f(a')$ then $a = a'$
Surjective (onto)	every point of the codomain is produced by some input	every seat is filled (some may be shared)	for every b there is an a with $f(a) = b$
Bijective	both injective and surjective	every seat has exactly one person	every b has exactly one a

Worked examples with small sets

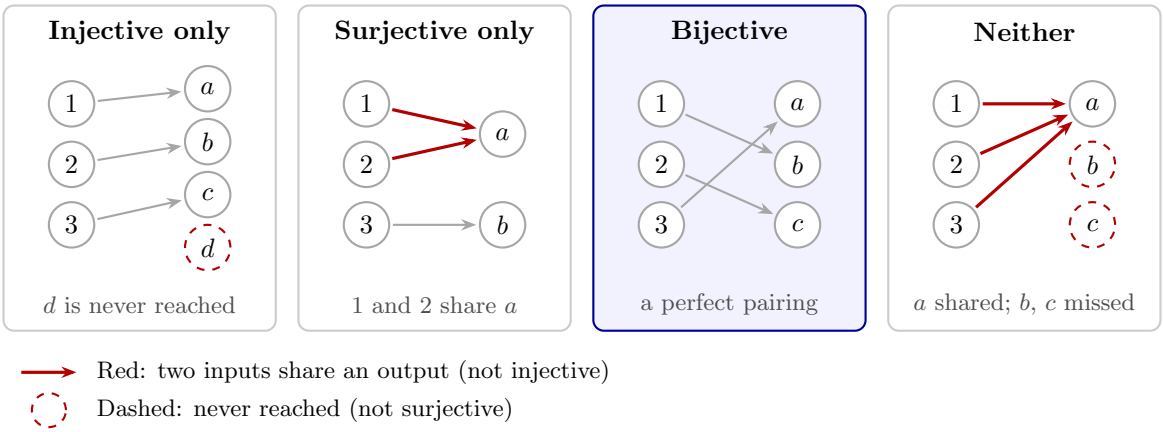
Let the inputs be $\{1, 2, 3\}$.

1. Outputs $\{a, b, c, d\}$; $f(1) = a$, $f(2) = b$, $f(3) = c$. **Injective, not surjective.** No two

inputs share an output, but d is never reached.

2. Outputs $\{a, b\}$; $f(1) = a$, $f(2) = a$, $f(3) = b$. **Surjective, not injective.** Both a and b are reached, but 1 and 2 share a.
3. Outputs $\{a, b, c\}$; $f(1) = b$, $f(2) = c$, $f(3) = a$. **Bijective.** A perfect pairing.
4. Outputs $\{a, b, c\}$; $f(1) = a$, $f(2) = a$, $f(3) = a$. **Neither.** Everyone shares a, and b and c are never reached.

The picture below draws all four.



Red arrows break injectivity; dashed outputs break surjectivity. Only the bijection has neither problem.

A counting observation (it comes back in Section 9.6). With 3 people and 4 seats, you can be injective but never surjective: there are not enough people to fill every seat. With 3 people and 2 seats, you can be surjective but never injective: someone must share. **A bijection needs exactly as many inputs as outputs.** For matrices, this is why only square matrices can be invertible.

Worked examples with real functions (inputs and outputs are all real numbers $\mathbb{R}$). A handy A-level tool is the **horizontal line test**: draw horizontal lines $y = c$ across the graph. If every such line hits the graph at most once, the function is injective. If every line hits it at least once, it is surjective.

Function	Injective?	Surjective?	Reason
$f(x) = 2x + 1$	Yes	Yes	any y comes from exactly one x: $x = (y - 1)/2$
$f(x) = x^3$	Yes	Yes	always increasing; y comes from $x = \sqrt[3]{y}$
$f(x) = e^x$	Yes	No	always increasing, but never negative: nothing gives -1
$f(x) = x^2$	No	No	$f(-1) = f(1) = 1$, and -1 is never produced

The codomain matters. x^2 from $\mathbb{R}$ to $[0, \infty)$ (only non-negative outputs allowed) is surjective, because every non-negative number is a square. If we also restrict the inputs to $[0, \infty)$, it becomes injective too, so it is a bijection with inverse $\sqrt{x}$. Same rule, different sets, different answer. For matrices there is no such choice: the shape of the matrix fixes both sets.

9.4 An inverse exists exactly when the map is bijective

A function g is an **inverse** of f if it undoes f and f undoes it:

$$g(f(a)) = a \text{ for every input } a, \qquad f(g(b)) = b \text{ for every } b \text{ in the codomain.}$$

The claim: f has an inverse $\iff$ f is bijective.

Why a bijection can be undone. Every output b has exactly one input; let $g(b)$ be that input.

Worked example

For the bijection $f(1) = b$, $f(2) = c$, $f(3) = a$, the undo machine is $g(a) = 3$, $g(b) = 1$, $g(c) = 2$. Check: $g(f(1)) = g(b) = 1$, as required.

Why anything else cannot be undone.

- **Not injective → stuck.** If two inputs share an output, say $f(1) = f(2) = a$, what should $g(a)$ be? It must be 1 (to undo $f(1)$) and also 2 (to undo $f(2)$). Impossible. In symbols: if $f(a) = f(a')$, apply g to both sides to get $a = a'$.
- **Not surjective → stuck.** If d is never produced, then $f(g(d)) = d$ cannot hold, because f never outputs d. In symbols: for any b, the input $a = g(b)$ satisfies $f(a) = b$, so every b is reached.

For matrices. The two conditions $A^{-1}A = I$ and $AA^{-1} = I$ say exactly that $w \mapsto A^{-1}w$ is the undo machine of $v \mapsto Av$. Conversely, when $v \mapsto Av$ is a bijection, its undo machine turns out to also be a matrix (a fact we accept here). So:

$$A \text{ is invertible} \iff v \mapsto Av \text{ is bijective.}$$

Worked example

For $A = \left[\begin{smallmatrix} 2 & 3 \\ 3 & 5 \end{smallmatrix}\right]$, solving $Av = w$ always has exactly one answer, $v = A^{-1}w$. With $w = (1, 0)$: $v = \left[\begin{smallmatrix} 5 & -3 \\ -3 & 2 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right] = (5, -3)$. Check: $A(5, -3) = (10 - 9,\ 15 - 15) = (1, 0)$.

9.5 Injectivity is decided by the null space

Two sets attached to a matrix answer the two halves of the bijection question:

$$\text{Null}(A) = \{v : Av = 0\}, \qquad \text{Col}(A) = \{Av : v \text{ any input}\}.$$

- The **null space** $\text{Null}(A)$: all inputs the machine turns into the zero vector. It always contains $v = 0$, since $A0 = 0$.
- The **column space** $\text{Col}(A)$: all outputs the machine can produce. Since

  $$Av = v_1(\text{column } 1) + \cdots + v_p(\text{column } p),$$

  these are exactly the linear combinations of the columns. The rank is the number of independent directions in this set.

Worked example 1

$A = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right]$.

- Null space: $Av = 0$ means $v_1 + 2v_2 = 0$ (the second row says the same thing twice over).

So $v_1 = -2v_2$, and the null space is all multiples of $(-2, 1)$: a whole line.

- Column space: $Av = (v_1 + 2v_2,\ 2v_1 + 4v_2) = (v_1 + 2v_2) \times (1, 2)$. Every output is a multiple of $(1, 2)$: another line.

Worked example 2

$A = \left[\begin{smallmatrix} 2 & 3 \\ 3 & 5 \end{smallmatrix}\right]$. Null space: $2v_1 + 3v_2 = 0$ gives $v_1 = -1.5v_2$. Then the second row: $3(-1.5v_2) + 5v_2 = 0.5v_2 = 0$, so $v_2 = 0$ and $v_1 = 0$. The null space is just $\{0\}$.

The key trick for matrices. Because $A(v - w) = Av - Aw$:

$$Av = Aw \iff A(v - w) = 0 \iff v - w \in \text{Null}(A).$$

Two different inputs collide exactly when their difference is a non-zero vector in the null space.

Worked example: a collision

With $A = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right]$, take $v = (1, 0)$ and $w = (-1, 1)$. Then $Av = (1, 2)$ and $Aw = (-1 + 2,\ -2 + 4) = (1, 2)$: a collision. And $v - w = (2, -1)$ is in the null space, since $A(2, -1) = (0, 0)$.

So the injectivity question reduces to one question about zero:

$$\begin{aligned} v \mapsto Av \text{ is injective } &\iff \text{Null}(A) = \{0\} \iff Av = 0 \text{ only for } v = 0 \\ &\iff \text{the columns of } A \text{ are linearly independent} \iff \text{rank}(A) = p. \end{aligned}$$

Matrices collapse whole lines, not just pairs. If $Au = 0$ for some $u \neq 0$, then for any number t:

$$A(v + tu) = Av + tAu = Av.$$

So every point on the line through v in the direction u lands on the same output. The function x^2 merges just pairs (-1 and 1); a matrix that fails to be injective merges entire lines. Whatever part of the input lay along u is destroyed and cannot be recovered. In OLS, this is exactly when β and $\beta + tv$ give the same predictions.

9.6 Rank–nullity: why square matrices are special

First, **dimension** means "number of independent directions": a line has dimension 1, a flat plane 2, and the set $\{0\}$ (just one point) has dimension 0. The **rank–nullity theorem** says that for an $m \times p$ matrix A:

$$\dim \text{Null}(A) + \text{rank}(A) = p.$$

Read it as a budget. There are p input directions. Each one is either collapsed (it goes into the null space) or survives into the outputs (it counts towards the rank). Every collapsed direction is one direction missing from the outputs.

Worked example checks

$\left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right]$: null space is a line (1) and column space is a line (1), and $1 + 1 = 2$.
$\left[\begin{smallmatrix} 2 & 3 \\ 3 & 5 \end{smallmatrix}\right]$: null space $\{0\}$ (0) and rank 2, and $0 + 2 = 2$.

Square matrices ($m = p$). If nothing is collapsed, the rank is p, so the outputs fill all p dimensions: surjective as well, hence bijective, hence invertible. If something is collapsed, the rank is below p, so some outputs are missed. For a square matrix, injective, surjective and

invertible stand or fall together, so you only need to check whether $Av = 0$ has a non-zero solution. (Compare the counting observation in Section 9.3: same number of seats as people.)

Tall matrices ($m > p$), like a design matrix with $n > p$. The rank is at most p, which is less than n, so the outputs can never fill all n dimensions. X can never be surjective, so X never has an inverse, whatever its numbers. The best it can be is injective.

Worked example: why least squares is needed at all

For the running example $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, every output is $X\beta = (\beta_0 + \beta_1,\ \beta_0 + 2\beta_1,\ \beta_0 + 3\beta_1)$. These three numbers are always evenly spaced, so the middle one is the average of the other two. Our data $y = (1, 3, 2)$ is not like that: $3 \neq (1 + 2)/2$. So no β gives $X\beta = y$ exactly. The best we can do is get as close as possible: least squares.

This is the real reason OLS works with $X^\mathsf{T}X$. The tall X can never be inverted, but $X^\mathsf{T}X$ is square, and for a square matrix "injective" is enough to guarantee "invertible".

9.7 Where semidefiniteness fits in

We know $X^\mathsf{T}X \succeq 0$ always (Chapter 7), and that definiteness fails only when some $v \neq 0$ has $Xv = 0$. In fact X and $X^\mathsf{T}X$ collapse **exactly the same** directions:

- If $Xv = 0$, then $X^\mathsf{T}Xv = X^\mathsf{T}0 = 0$.
- If $X^\mathsf{T}Xv = 0$, then $v^\mathsf{T}X^\mathsf{T}Xv = 0$, which says $\|Xv\|^2 = 0$, and a zero length means $Xv = 0$.

$$\text{Null}(X^\mathsf{T}X) = \text{Null}(X).$$

Both matrices have p columns, so by rank–nullity they have the same rank: $\text{rank}(X^\mathsf{T}X) = p - \dim \text{Null}(X^\mathsf{T}X) = p - \dim \text{Null}(X) = \text{rank}(X)$. That proves the fact promised in Section 8.3.

The number $v^\mathsf{T}X^\mathsf{T}Xv = \|Xv\|^2$ measures **how much length the direction v keeps** after passing through X:

- Positive semidefinite: no direction ends up with negative length. Always true.
- Positive definite: no non-zero direction is squashed to length zero. X collapses nothing.

Key idea

Positive definite = positive semidefinite + nothing collapsed. For the square matrix $X^\mathsf{T}X$, "nothing collapsed" is exactly injectivity, and by rank–nullity injectivity is exactly invertibility.

9.8 A visual picture: why the dependent example is not bijective

Return to the dependent $X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$ from Chapter 8. For an input $v = (a, b)$:

$$X^\mathsf{T}X = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}, \qquad X^\mathsf{T}X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 14a + 28b \\ 28a + 56b \end{bmatrix} = 14(a + 2b) \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The output depends on a and b **only through the single number** $a + 2b$. Feed in a few inputs and watch:

Input (a, b)	$a + 2b$	Output
$(1, 0)$	1	$(14, 28)$
$(-1, 1)$	1	$(14, 28)$
$(3, -1)$	1	$(14, 28)$
$(0, 1)$	2	$(28, 56)$
$(-2, 1)$	0	$(0, 0)$

- **Not injective.** The first three inputs all give $(14, 28)$. Every input on the line $a + 2b = 1$ does. The difference between any two of them, such as $(1, 0) - (-1, 1) = (2, -1)$, lies in the null space, the line through $(-2, 1)$.
- **Not surjective.** Every output is a multiple of $(1, 2)$, so outputs fill only one line in the plane (rank 1). For example, $X^\mathsf{T}Xv = (1, 0)$ has no solution: it would need $14(a + 2b) = 1$ and $28(a + 2b) = 0$ at the same time.
- **The budget.** $2 = 1 + 1$: one input direction is collapsed, so only one survives.

So the equation $X^\mathsf{T}Xv = b$ has either no solution or infinitely many, never exactly one. An inverse would have to return *the* solution, so $(X^\mathsf{T}X)^{-1}$ cannot exist. By contrast, Example A in Chapter 11 has $X^\mathsf{T}X = \left[\begin{smallmatrix} 2 & 3 \\ 3 & 5 \end{smallmatrix}\right]$, which collapses nothing: it is a bijection and its inverse exists.

9.9 Summary

For a design matrix X $(n \times p)$, all of the following say the same thing:

1. The columns of X are linearly independent.
2. $Xv = 0$ only for $v = 0$.
3. $\beta \mapsto X\beta$ is injective: different coefficients always give different predictions.
4. $\text{rank}(X) = p$.
5. $\text{Null}(X^\mathsf{T}X) = \{0\}$.
6. $X^\mathsf{T}X$ is bijective, that is, invertible.
7. $X^\mathsf{T}X$ is positive definite.

When they fail, the cause is always the same: a non-zero direction v with $Xv = 0$, which X and $X^\mathsf{T}X$ both collapse.

10 What this means for OLS: uniqueness and geometry

The OLS score is always a bowl or a trough: any flat point is a lowest point, and the lowest point is unique exactly when X has full column rank.

The score can be written as a squared length:

$$S(\beta) = \|y - X\beta\|^2 .$$

The second derivative tells you the shape. At A-level you used $f''(x) > 0$ to show a turning point is a minimum. With several parameters, the second derivatives form a table called the

Hessian: entry (j, k) is $\partial^2 S/\partial\beta_j\partial\beta_k$ (differentiate by β_j, then by β_k). For OLS it is always

$$\nabla^2_\beta S = 2X^\mathsf{T}X.$$

Worked example: the Hessian

From Chapter 5, $S = 14 - 12\beta_0 - 26\beta_1 + 3\beta_0^2 + 12\beta_0\beta_1 + 14\beta_1^2$. Its second derivatives are $\partial^2 S/\partial\beta_0^2 = 6$, $\partial^2 S/\partial\beta_0\partial\beta_1 = 12$ and $\partial^2 S/\partial\beta_1^2 = 28$. So the Hessian is $\left[\begin{smallmatrix} 6 & 12 \\ 12 & 28 \end{smallmatrix}\right]$, which is $2 \times \left[\begin{smallmatrix} 3 & 6 \\ 6 & 14 \end{smallmatrix}\right] = 2X^\mathsf{T}X$.

Convex means "no false bottoms". Since $X^\mathsf{T}X \succeq 0$ (Chapter 7), the Hessian is never negative in any direction, so S never curves downwards anywhere. A function like this is called **convex**: it has no bumps and no separate dips, so any point where the gradient is zero is a lowest point overall (a global minimum). This is why setting the gradient to zero in Chapter 5 was enough; we never need to check for maxima or saddle points.

Now the two cases, matching the bowl and the trough from Chapter 6:

- X **has full column rank → a bowl.** $X^\mathsf{T}X$ is positive definite, so S curves upwards in every direction. There is exactly one lowest point: $\hat{\beta}$ is unique.
- X **is not full column rank → a trough.** There is a $v \neq 0$ with $Xv = 0$. Sliding β along v changes nothing:

  $$X(\beta + tv) = X\beta + tXv = X\beta.$$

  The floor of the trough is a whole line of equally good coefficient vectors. The **predictions** $X\hat{\beta}$ are still unique (every point on the floor gives the same ones); only the **coefficients** are not.

Worked example

With the dependent $X = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{smallmatrix}\right]$, the coefficients $\beta = (1, 0)$ and $\beta = (-1, 1)$ give the same predictions: $X(1, 0) = (1, 2, 3)$ and $X(-1, 1) = (-1 + 2,\ -2 + 4,\ -3 + 6) = (1, 2, 3)$. Their difference $(2, -1)$ is in the null space.

10.1 Eigenvalue intuition

What is an eigenvector? Most vectors get turned to a new direction when you multiply them by a matrix. An **eigenvector** is a special non-zero vector that only gets stretched or shrunk, keeping its direction. The stretch factor λ ("lambda") is its **eigenvalue**:

$$Av = \lambda v.$$

Worked example 1

$A = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$.

- $v = (1, 1)$: $Av = (2, 2) = 2v$. Eigenvalue 2: this direction is doubled.
- $v = (1, -1)$: $Av = (0, 0) = 0v$. Eigenvalue 0: this direction is **collapsed to nothing**.

Worked example 2: the dependent $X^\mathsf{T}X$

$X^\mathsf{T}X = \left[\begin{smallmatrix} 14 & 28 \\ 28 & 56 \end{smallmatrix}\right]$.

- $v = (1, 2)$: $X^\mathsf{T}Xv = (14 + 56,\ 28 + 112) = (70, 140) = 70v$. Eigenvalue 70.

- $v = (-2, 1)$: $X^\mathsf{T}Xv = (-28+28,\ -56+56) = (0, 0)$. Eigenvalue 0, along the null-space direction from Chapter 8.

The connections:

- An eigenvalue of 0 means a non-zero v with $Av = 0$, so a collapsed direction, so A is singular. **A symmetric matrix is invertible exactly when none of its eigenvalues is 0.**
- For a positive semidefinite matrix, eigenvalues are never negative. Quick proof: if $Av = \lambda v$, then $v^\mathsf{T}Av = \lambda v^\mathsf{T}v = \lambda \|v\|^2$. The left side is ≥ 0 and $\|v\|^2 > 0$, so $\lambda \geq 0$.
- For the Hessian $2X^\mathsf{T}X$, an eigenvalue says **how steeply the bowl curves** in that eigenvector's direction. An eigenvalue of 0 is a perfectly flat direction: the floor of the trough. It appears exactly when the columns of X are dependent.

11 Worked examples

Example A has independent columns and one exact answer; Example B has identical columns and infinitely many equally good answers.

11.1 Example A: full column rank and a unique solution

A straight-line model with intercept and slope, fitted to two points: $(x, y) = (1, 2)$ and $(2, 3)$.

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Step 1: $X^\mathsf{T}X$. The columns are $c_1 = (1, 1)$ and $c_2 = (1, 2)$. Dot them in every pairing: $c_1 \cdot c_1 = 2$, $c_1 \cdot c_2 = 1 + 2 = 3$, $c_2 \cdot c_2 = 1 + 4 = 5$.

$$X^\mathsf{T}X = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}.$$

(Here X happens to be symmetric, so $X^\mathsf{T} = X$. That is a coincidence of this example, not a rule.)

Step 2: is it invertible? The determinant is $2 \times 5 - 3 \times 3 = 1$, which is not 0. So yes.

Step 3: the inverse. Swap the diagonal entries, flip the signs of the others, divide by the determinant (1):

$$(X^\mathsf{T}X)^{-1} = \frac{1}{1}\begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}.$$

We checked in Section 9.1 that this really multiplies to I.

Step 4: $X^\mathsf{T}y$. Each column dotted with y: $c_1 \cdot y = 2+3 = 5$, $c_2 \cdot y = 2+6 = 8$. So $X^\mathsf{T}y = (5, 8)$.

Step 5: $\hat{\beta}$.

$$\hat{\beta} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 25 - 24 \\ -15 + 16 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The fitted line is $\hat{y} = 1 + x$.

Step 6: check. Predictions: $X\hat{\beta} = (1+1,\ 1+2) = (2,3) = y$. The residuals are all zero. That makes sense: with two points and two parameters, a straight line can always pass through both exactly. Also check the normal equations: $\left[\begin{smallmatrix}2&3\\3&5\end{smallmatrix}\right]\left[\begin{smallmatrix}1\\1\end{smallmatrix}\right] = \left[\begin{smallmatrix}5\\8\end{smallmatrix}\right] = X^{\mathsf{T}}y$.

The two columns of X are independent, so the coefficients are uniquely pinned down.

11.2 Example B: dependent columns and no inverse

Now both observations have $x = 1$, and suppose the outcomes were $y = (1,3)$.

$$X = \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, \qquad y = \begin{bmatrix}1\\3\end{bmatrix}.$$

The two columns are identical: the feature never varies, so it looks exactly like the intercept.

Step 1: $X^{\mathsf{T}}X$ and its determinant.

$$X^{\mathsf{T}}X = \begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}, \qquad \det(X^{\mathsf{T}}X) = 2\times 2 - 2\times 2 = 0.$$

Singular: no inverse. The formula $\hat{\beta} = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$ cannot be used.

Step 2: solve the normal equations anyway. $X^{\mathsf{T}}y = (1+3,\ 1+3) = (4,4)$. The normal equations are $2\beta_0 + 2\beta_1 = 4$, written twice. That is really just one equation: $\beta_0 + \beta_1 = 2$.

Step 3: list the solutions. For any number t,

$$\begin{bmatrix}\beta_0\\ \beta_1\end{bmatrix} = \begin{bmatrix}1+t\\ 1-t\end{bmatrix} \quad \Rightarrow \quad \beta_0 + \beta_1 = 2.$$

For example $(1,1)$, $(2,0)$, $(0,2)$ and $(-5,7)$ all work.

Step 4: they all predict the same thing. Every solution predicts $\beta_0 + \beta_1 = 2$ for both observations: the average of 1 and 3, which is the best single guess. The residuals are $(-1,1)$ and $S = 2$ every time.

So a best fit exists and the predictions are unique; the problem is only that the coefficients cannot be separated. The data contains no information about how to split the total of 2 between the intercept and the slope.

11.3 Try it yourself

Try it yourself

1. Fit a line to the points $(0,1)$, $(1,2)$, $(2,4)$ using $\hat{\beta} = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}y$.
2. Does $X = \left[\begin{smallmatrix}1&2\\1&2\\1&2\end{smallmatrix}\right]$ have full column rank? If not, find a non-zero v with $Xv = 0$.
3. Is $f(x) = 3x - 4$ (from $\mathbb{R}$ to $\mathbb{R}$) injective, surjective, bijective? What about $f(x) = |x|$?
4. Is $A = \left[\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right]$ invertible? If so, find A^{-1} and check that $AA^{-1} = I$.

Answers

1. $X = \left[\begin{smallmatrix}1&0\\1&1\\1&2\end{smallmatrix}\right]$, $X^{\mathsf{T}}X = \left[\begin{smallmatrix}3&3\\3&5\end{smallmatrix}\right]$ (determinant 6), $X^{\mathsf{T}}y = (7,10)$. Then $\hat{\beta} = \frac{1}{6}\left[\begin{smallmatrix}5&-3\\-3&3\end{smallmatrix}\right]\left[\begin{smallmatrix}7\\10\end{smallmatrix}\right] =$

$\frac{1}{6}(5, 9) = (5/6,\ 1.5)$. The line is $\hat{y} = 5/6 + 1.5x$.

2. No: column 2 is twice column 1, so the rank is 1. $v = (2, -1)$ gives $Xv = 0$. (Check: $X^\mathsf{T}X = \left[\begin{smallmatrix} 3 & 6 \\ 6 & 12 \end{smallmatrix}\right]$, determinant 0.)
3. $3x - 4$ is bijective, with inverse $x = (y + 4)/3$. $|x|$ is neither: $|-1| = |1|$, and -1 is never produced.
4. Determinant $1 \times 4 - 2 \times 3 = -2 \neq 0$, so yes. $A^{-1} = \frac{1}{-2}\left[\begin{smallmatrix} 4 & -2 \\ -3 & 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} -2 & 1 \\ 1.5 & -0.5 \end{smallmatrix}\right]$. Check: the first row of AA^{-1} is $(-2 + 3,\ 1 - 1) = (1, 0)$ and the second is $(-6 + 6,\ 3 - 2) = (0, 1)$.

12 Common questions and corrections

The four questions students ask most, each answered with the examples from earlier chapters.

12.1 Does X^T need to be invertible?

No. In real data X is tall, for example 1,000 people $\times$ 5 features. Chapter 9 showed only square matrices can be invertible (a bijection needs as many inputs as outputs), so neither X nor X^T has an inverse. The formula only needs the square $p \times p$ matrix $X^\mathsf{T}X$ to be invertible.

A common mistake is to "simplify" $(X^\mathsf{T}X)^{-1}$ into $X^{-1}(X^\mathsf{T})^{-1}$. That would need X^{-1} to exist, which it does not.

12.2 If $X^\mathsf{T}X$ is singular, does OLS fail?

Not really. A best fit still exists, and its predictions are unique (Chapter 10). What fails is uniqueness of the coefficients: a whole line of β's is equally good.

One standard fix is the **Moore–Penrose pseudoinverse**, written X^+. Among all the equally good β's, it picks the one with the smallest length:

$$\hat{\beta} = X^+ y.$$

Worked example

In Example B, the solutions were $(1 + t,\ 1 - t)$. Their squared length is $(1 + t)^2 + (1 - t)^2 = 2 + 2t^2$, smallest at $t = 0$. So the pseudoinverse picks $\hat{\beta} = (1, 1)$.

In practice, the better fix is usually to remove the redundant feature (for example, drop one of the dummy columns in the dummy variable trap).

12.3 Why not always calculate the inverse directly?

Computers store numbers with limited precision, so tiny rounding errors creep into every calculation. Some problems amplify those errors badly; they are called **ill-conditioned**. The **condition number** measures how much a problem can amplify errors.

This happens when columns are *nearly* dependent, for example height in centimetres alongside height in inches rounded to the nearest inch. $X^\mathsf{T}X$ is technically invertible but only just, like dividing by 0.0000001.

Forming $X^\mathsf{T}X$ **squares** the condition number of X: a problem that amplifies errors 1,000 times becomes one that amplifies them 1,000,000 times. So numerical software avoids computing $(X^\mathsf{T}X)^{-1}$ at all. It uses methods called **QR factorisation** or **singular value decomposition (SVD)** that work with X directly. For example, NumPy's `np.linalg.lstsq` solves least squares this way. The formula $\hat{\beta} = (X^\mathsf{T}X)^{-1}X^\mathsf{T}y$ is for understanding, not for computing.

12.4 What about ridge regression?

Ridge regression adds a penalty for large coefficients, with a tuning number $\lambda > 0$:

$$\|y - X\beta\|^2 + \lambda \|\beta\|^2 .$$

Redoing Chapter 5 with the extra term gives normal equations with $X^\mathsf{T}X + \lambda I$ in place of $X^\mathsf{T}X$. This matrix is **always** positive definite, because for any $v \neq 0$:

$$v^\mathsf{T}(X^\mathsf{T}X + \lambda I)v = \|Xv\|^2 + \lambda \|v\|^2 > 0.$$

The first term is at least 0 and the second is strictly positive ($\lambda > 0$ and $v \neq 0$). So $X^\mathsf{T}X + \lambda I$ is invertible even when X has dependent columns.

Worked example

Example B with $\lambda = 1$: $X^\mathsf{T}X + I = \left[\begin{smallmatrix} 3 & 2 \\ 2 & 3 \end{smallmatrix}\right]$, determinant $9 - 4 = 5$, not 0. So

$$\hat{\beta} = \frac{1}{5}\begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.8 \end{bmatrix}.$$

One unique answer. The penalty pulls the coefficients towards zero (predictions 1.6 instead of 2), and in exchange it removes the flat trough.

(This is the plain version that also penalises the intercept; in practice the intercept is often left unpenalised.)

13 Compact derivative rule sheet

Six rules cover every derivative in OLS. Each one can be checked by writing out the case $p = 2$, as in Chapter 5.

Rule	In words	Check with $p = 2$
$\nabla c = 0$	anything with no β in it has zero gradient	$\nabla(14) = (0, 0)$
$\nabla(a^\mathsf{T}\beta) = a$	the gradient of a weighted sum is the list of weights	$\nabla(6\beta_0 + 13\beta_1) = (6, 13)$
$\nabla(\beta^\mathsf{T}a) = a$	same number as $a^\mathsf{T}\beta$, same gradient	as above
$\nabla(\beta^\mathsf{T}A\beta) = (A + A^\mathsf{T})\beta$	each β_k appears on both sides of A, so both count	$A = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right]$: $(2\beta_1 + 2\beta_2,\ 2\beta_1 + 6\beta_2)$
$A = A^\mathsf{T} \Rightarrow \nabla(\beta^\mathsf{T}A\beta) = 2A\beta$	the matrix version of $\frac{d}{d\beta}(a\beta^2) = 2a\beta$	$A = \left[\begin{smallmatrix} 3 & 6 \\ 6 & 14 \end{smallmatrix}\right]$: $(6\beta_0 + 12\beta_1,\ 12\beta_0 + 28\beta_1)$
$\nabla \|y - X\beta\|^2 = -2X^\mathsf{T}y + 2X^\mathsf{T}X\beta$	the OLS gradient: all the rules above combined	running example: $-2(6, 13) + 2X^\mathsf{T}X\beta$

$$\nabla_\beta \|y - X\beta\|^2 = -2X^\mathsf{T}y + 2X^\mathsf{T}X\beta.$$

Memory aid: each rule is the matrix twin of a one-variable rule. $a^\mathsf{T}\beta$ behaves like $a\beta$ (derivative a); $\beta^\mathsf{T}A\beta$ with symmetric A behaves like $a\beta^2$ (derivative $2a\beta$).

14 The full argument in six lines

The whole story, each step with its plain-English meaning:

1. **OLS minimises $S(\beta) = \|y - X\beta\|^2$.** Choose the coefficients that make the total squared error as small as possible.
2. **Differentiating with respect to the vector β means one partial derivative per coefficient, stacked into a gradient.** No new calculus, just ordinary derivatives in a list.
3. **Setting the gradient to zero gives the normal equations $X^\mathsf{T}X\hat{\beta} = X^\mathsf{T}y$.** At the bottom of the valley it is flat in every direction.
4. **For every v, $v^\mathsf{T}X^\mathsf{T}Xv = \|Xv\|^2 \geq 0$, so $X^\mathsf{T}X$ is positive semidefinite.** A squared length is never negative, so the valley is a bowl or a trough, never a saddle.
5. **It is positive definite, and hence invertible, exactly when $Xv = 0$ forces $v = 0$, that is, when X has full column rank.** No feature is a mixture of the others, so no direction is collapsed and the valley has a single lowest point.
6. **Then the unique solution is $\hat{\beta} = (X^\mathsf{T}X)^{-1}X^\mathsf{T}y$.** Undo $X^\mathsf{T}X$ to read off the answer.

Matrix notation is not a different calculus.
It is ordinary algebra and calculus written compactly.

Glossary

Term	Plain-English meaning	First seen
Scalar	a single number	1.1
Vector	an ordered list of numbers, usually a column	1.2
Matrix	a rectangular table of numbers, size rows × columns	1.3
Transpose (T)	swap rows and columns	1.2, 1.3
Linear combination	a sum of multiples of vectors	1.2
Design matrix X	the data: one row per observation, one column per feature	1.3
Dot product	multiply matching entries and add up	2.2
Partial derivative	rate of change in one parameter, others held fixed	3.2
Gradient	all partial derivatives of a single-number function, stacked	3.2
Jacobian	the table of derivatives of a list-valued function	3.3
Residual	truth minus prediction, $y_i - \hat{y}_i$	4
Symmetric matrix	equal to its own transpose	5.3
Normal equations	$X^\mathsf{T}X\hat{\beta} = X^\mathsf{T}y$: "gradient equals zero" for OLS	5.5
Determinant (2×2)	$ad - bc$; zero means no inverse	5.6
Positive semidefinite	$v^\mathsf{T}Av \geq 0$ for every v: a bowl or a trough	6.1
Positive definite	$v^\mathsf{T}Av > 0$ for every non-zero v: a bowl	6.1

Term	Plain-English meaning	First seen
Linearly independent	no vector is a mixture of the others	8.1
Rank	number of independent columns	8.1
Full column rank	rank equals the number of columns	8.1
Identity matrix I	the "do nothing" matrix, the matrix version of 1	9.1
Inverse A^{-1}	the "undo" matrix: $AA^{-1} = A^{-1}A = I$	9.1
Singular	square with no inverse	9.1
Domain, codomain	where inputs come from; where outputs are allowed to land	9.2
Injective	different inputs always give different outputs	9.3
Surjective	every point of the codomain is produced	9.3
Bijective	both: a perfect pairing of inputs and outputs	9.3
Null space	all inputs sent to zero	9.5
Column space	all possible outputs	9.5
Rank–nullity	collapsed directions + surviving directions = number of columns	9.6
Hessian	the table of second derivatives; $2X^\top X$ for OLS	10
Convex	curves upwards or stays flat everywhere; no false bottoms	10
Eigenvalue	how much a matrix stretches its special (eigenvector) direction	10.1
Pseudoinverse X^+	picks the shortest β among equally good ones	12.2
Ridge regression	OLS plus a penalty $\lambda \|\beta\|^2$ that guarantees one answer	12.4